

```
oStatusLine.draw()

pygame.display.update()
clock.tick(FRAMES_PER_SECOND)
```

---

*Listing 9-5: The main program that creates random shapes from three classes*

As we saw in Chapter 4, whenever we have a large number of objects to manage, the typical approach is to build a list of objects. So, before the main loop starts, the program first builds a list of shapes ❶ by randomly choosing among a circle, a square, and a triangle; creating an object of that type; and appending it to the list. Using this approach, we can then iterate over the list and call methods of the same name in every object in the list.

Inside the main loop, the program checks for the mouse down event ❷ that happens when the user clicks. Whenever the event is detected, the code iterates through the `shapesList` ❸ and calls the `clickedInside()` ❹ method for each shape. Because of polymorphism, it doesn't matter which class the object was instantiated from. Again, the key is that the implementation of the `clickedInside()` method can be different for different classes.

When any `clickedInside()` method returns `True` ❺, we call the `getArea()` then `getType()` methods of that object, without worrying about which type of object was clicked on.

Here is the output of a typical run, after clicking on a few of the different shapes:

---

```
Clicked on a Circle whose area is 5026.544
Clicked on a Square whose area is 1600
Clicked on a Triangle whose area is 1982.5
Clicked on a Square whose area is 1600
Clicked on a Square whose area is 100
Clicked on a Triangle whose area is 576.0
Clicked on a Circle whose area is 3019.06799
```

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